Previous YFT Assessment from 2023

Data, Model, Results, Innovations, Issues

Arni Magnusson

SPC PAW Meeting, Nouméa
24 March 2026

Overview

Data and Model *model fit to data, structural uncertainty*

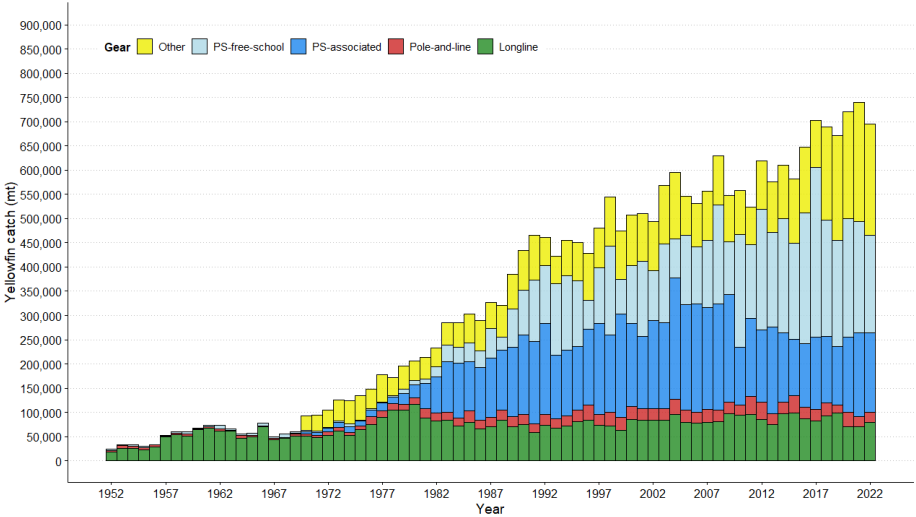
Results *reference points, stock status*

Innovations *simplified regional structure, Lorenzen M, catch-conditioned, estimation uncertainty*

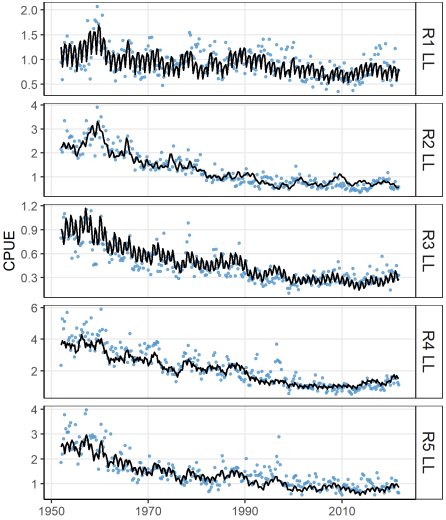
Issues *convergence/jittering, narrow confidence intervals, likelihood conflict between length and weight comps*

Data and Model

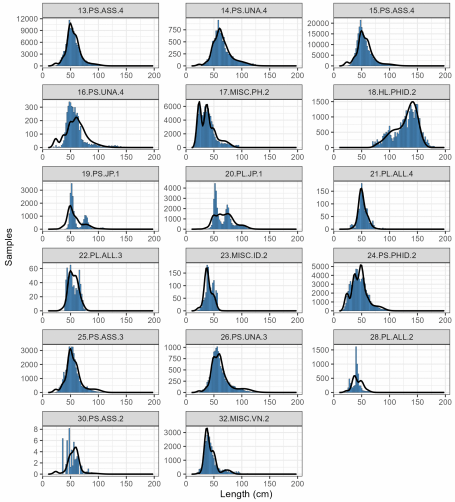
Catches

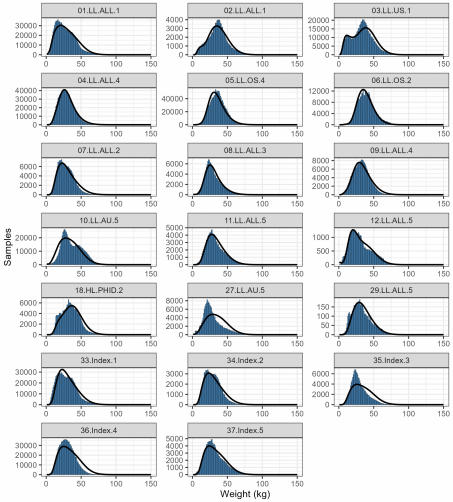

Gear
Other
PS-free-school
PS-associated
Pole-and-line
Longline
Yellowfin catch (mt)
900,000
850,000
800,000
750,000
700,000
650,000
600,000
550,000
500,000
450,000
400,000
350,000
300,000
250,000
200,000
150,000
100,000
50,000
0
1952
1957
1962
1967
1972
1977
1982
1987
1992
1997
2002
2007
2012
2017
2022
Year

CPUE

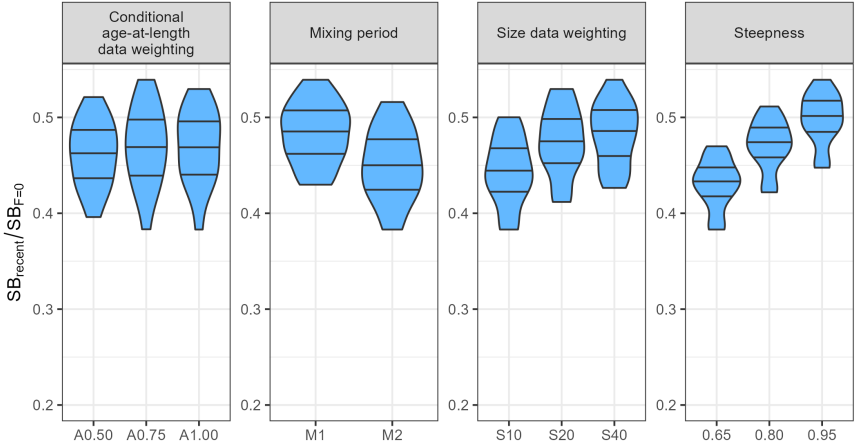

CPUE
R1 LL
R2 LL
R3 LL
R4 LL
R5 LL
2.0
1.5
1.0
0.5
4
3
2
1
1.2
0.9
0.6
0.3
6
4
2
4
3
2
1
1950
1970
1990
2010

Size composition



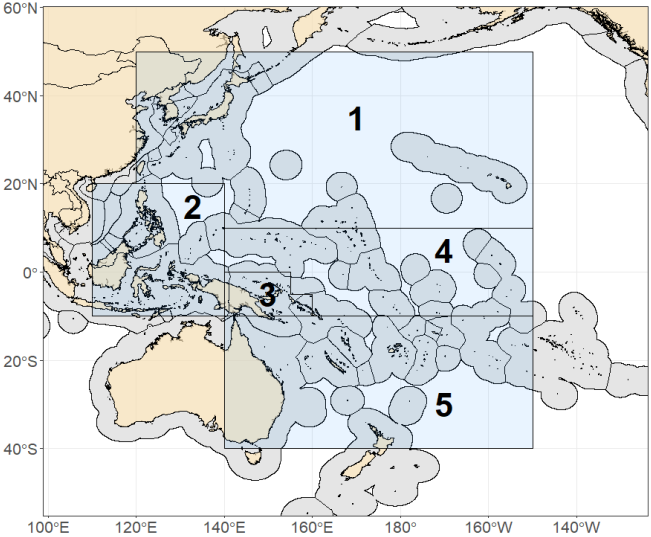


Structural uncertainty

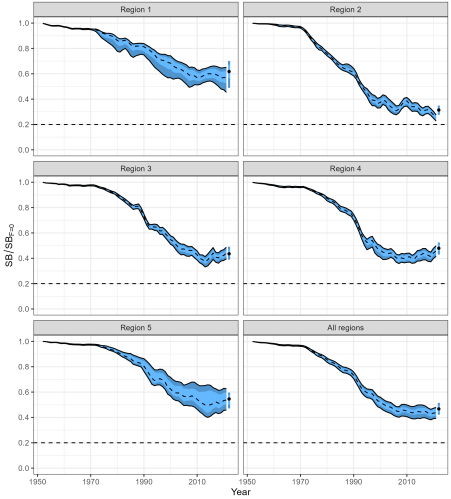


2023 Assessment Results

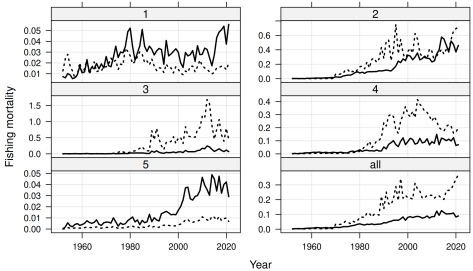
Regions

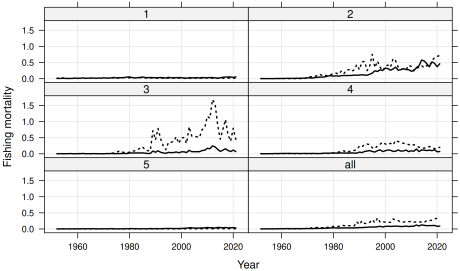


Stock status

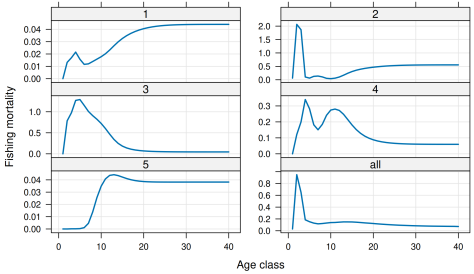


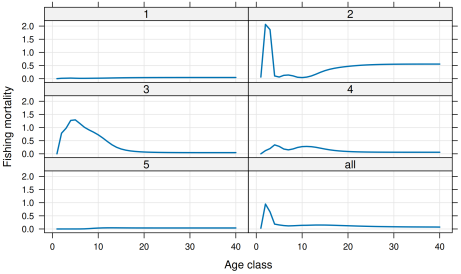
Fishing mortality: juvenile and adult



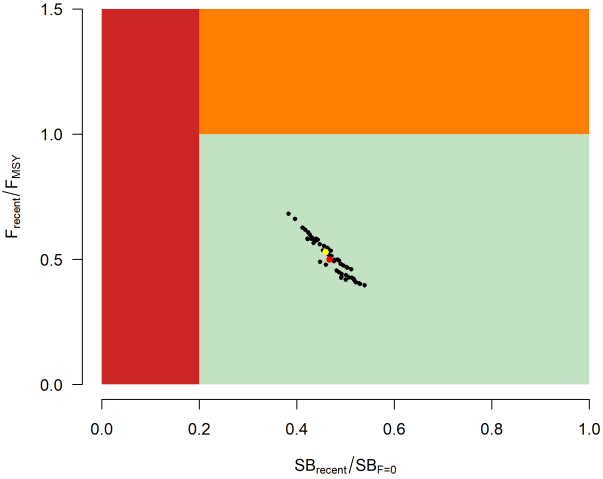


Fishing mortality: last ten years



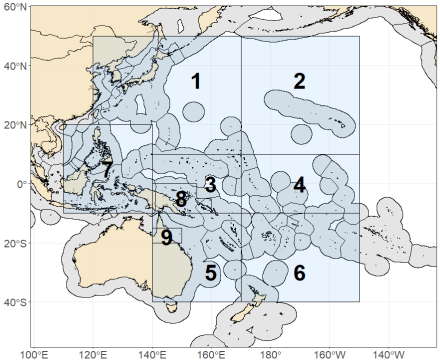


Majuro plot of stock status

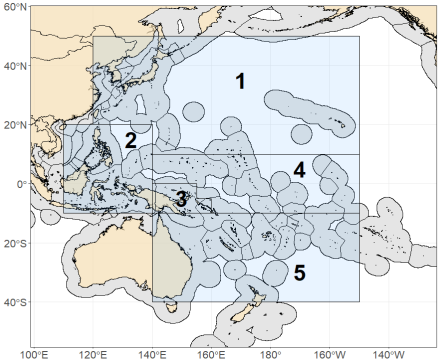


Innovations in 2023 Assessment

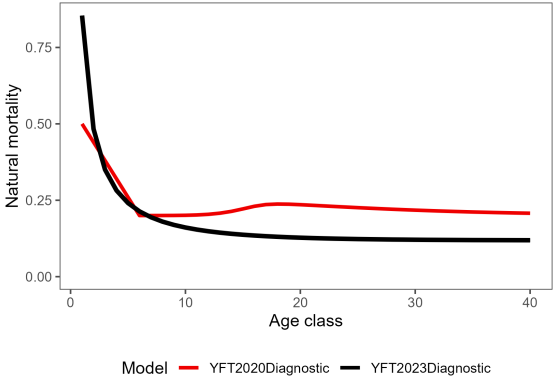
Simplified regional structure



⇒



Lorenzen natural mortality



Catch-conditioned method

MULTIFAN-CL User Manual, section 4.5:

Essentially the catch-conditioned method for estimating fishing mortality, and hence catch, is that employed in most contemporary fish population models, such that the fishing mortalities are solved for conditional upon the observed catches. The formulation used in this method assumes that the Baranov catch equation held, and the Newton-Raphson (N-R) procedure was used to solve for the fishing mortality levels among the various fisheries for which there were incidents within a time period.

There is a fundamental difference in the dynamic model calculations when using the catch-conditioned method compared to the catch-errors method. With the latter, all the estimated parameters for calculating fishing mortality (catchability, effort deviates etc.) are available at the beginning of the model calculation period for all time periods; and therefore, the numbers-at-age calculation is a linear sequence starting from the first model year. In contrast, for the catch-conditioned method, in the first model time period none of the fishing mortalities are available, but rather, they are calculated sequentially for each fishing incident with each time period, starting in the first year.

Estimation uncertainty

	Mean	Median	Min	10%ile	90%ile	Max	Diagnostic model
C_{latest}	751657	751856	750785	750860	752268	752337	751908
F_{MSY}	0.07	0.07	0.06	0.06	0.09	0.09	0.07
f_{mult}	1.96	2.00	1.47	1.64	2.38	2.50	1.89
$F_{\text{recent}}/F_{\text{MSY}}$	0.51	0.50	0.40	0.42	0.61	0.68	0.53
MSY	697874	700400	616800	644320	739560	771600	671600
SB_0	5761796	5729000	4455000	4817200	6640900	7279000	5216000
$SB_{F=0}$	5633743	5603267	4624645	4907798	6280841	6825888	5173954
SB_{latest}/SB_0	0.49	0.50	0.41	0.44	0.54	0.56	0.49
$SB_{\text{latest}}/SB_{F=0}$	0.50	0.50	0.41	0.45	0.55	0.58	0.49
$SB_{\text{latest}}/SB_{\text{MSY}}$	2.49	2.48	1.78	1.91	3.11	3.16	2.44
SB_{MSY}	1177733	1160500	740400	838260	1538200	1707000	1044000
SB_{MSY}/SB_0	0.20	0.20	0.17	0.17	0.23	0.24	0.20
$SB_{\text{MSY}}/SB_{F=0}$	0.21	0.21	0.16	0.17	0.24	0.25	0.20
$SB_{\text{recent}}/SB_{F=0}$	0.47	0.47	0.38	0.42	0.52	0.54	0.46
$SB_{\text{recent}}/SB_{\text{MSY}}$	2.31	2.30	1.68	1.77	2.89	2.94	2.27
$Y_{F_{\text{recent}}}$	157188	155300	141400	145150	172270	173300	152500
Including estimation uncertainty:							
	Mean	Median	Min	10%ile	90%ile	Max	
$SB_{\text{recent}}/SB_{F=0}$	0.47	0.47	0.36	0.42	0.52	0.59	
$F_{\text{recent}}/F_{\text{MSY}}$	0.51	0.50	0.26	0.41	0.62	0.78	
$SB_{\text{recent}}/SB_{\text{MSY}}$	2.31	2.28	0.93	1.73	2.95	3.59	

Issues in 2023 Assessment

Convergence and jittering

Table 6: Overview of grid models. Each grid model was jittered 20 times and the model run that had the best objective function value was selected. The columns show the final gradient, final dynamic depletion, number of negative eigenvalues, and whether the Hessian was positive definite.

	Grid model	Njitters	ObjFun	Gradient	Depletion	Neigen	PDH
1	`m1_s10_a050_h65`	20	-838059.98	0.00229	0.431	3	
2	`m1_s10_a050_h80`	20	-838060.01	0.00018	0.468	3	
3	`m1_s10_a050_h95`	20	-838060.02	0.00414	0.492	2	
4	`m1_s10_a075_h65`	20	-837227.54	0.01139	0.430	4	
5	`m1_s10_a075_h80`	20	-837227.56	0.01899	0.467	4	
6	`m1_s10_a075_h95`	20	-837227.58	0.00139	0.491	4	
7	`m1_s10_a100_h65`	20	-836390.85	0.00095	0.440	0	Yes
8	`m1_s10_a100_h80`	20	-836390.87	0.00016	0.476	0	Yes
9	`m1_s10_a100_h95`	20	-836390.89	0.00637	0.500	0	Yes
10	`m1_s20_a050_h65`	20	-747997.99	0.00398	0.455	3	
...	...	...	...	...	...	...	...
54	`m2_s40_a100_h95`	20	-649867.30	0.00279	0.516	3	

Narrow confidence intervals

- Incorporating estimation uncertainty had no real effect on the overall uncertainty. For example, the confidence interval for $SB_{\text{recent}}/SB_{F=0}$:

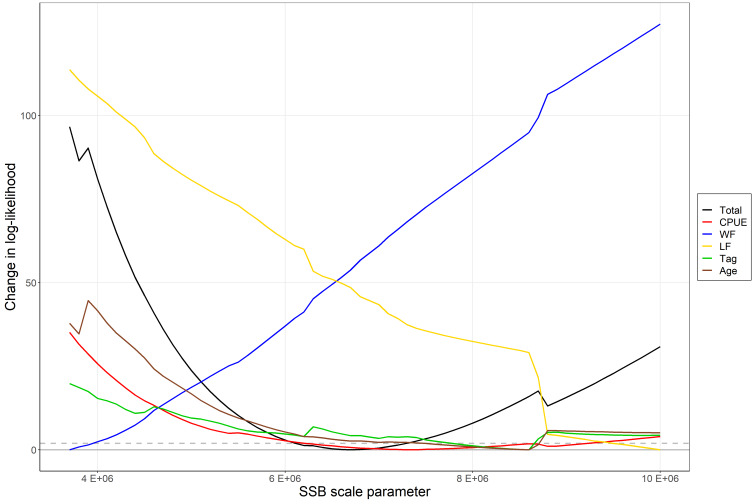
  0.42 − 0.52 without estimation uncertainty

  0.42 − 0.52 including estimation uncertainty

- Also, the confidence interval around the estimated M is quite narrow: 0.11 − 0.13

The Lorenzen scale parameter that is estimated in the current diagnostic model determines M at the highest age in the model. This parameter is estimated by the diagnostic model as 0.119 per quarter, with 95% confidence limits at 0.113 and 0.126, based on the Hessian.

Likelihood conflict between length and weight comps



Summary

Data and Model *model fit to data, structural uncertainty*

Results *reference points, stock status*

Innovations *simplified regional structure, Lorenzen M, catch-conditioned, estimation uncertainty*

Issues *convergence/jittering, narrow confidence intervals, likelihood conflict between length and weight comps*

These slides:

https://github.com/PacificCommunity/ofp-sam-yft-2026-assessment